

Engagement With a Coordination Game Decoupled From Fall Risk Assessment: Exploratory Observational Study

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Abstract

Background: Failure-induced abandonment syndrome (FIAS) has been proposed to describe withdrawal after a personally salient adverse experience with gamified fall risk assessment (FRA). A candidate explanation concerns appraisal of one's own capability. Removing clinical apparatus and outputs offers a way to investigate this question, although changes in game design, access and social context may also affect participation.

Objective: To describe engagement in a coordination game decoupled from FRA apparatus and outputs, and examine its implications for investigating FIAS and learning from gameplay experience.

Methods: Stage 3A was an exploratory observational deployment between April and July 2026. Twenty distributors were invited, each asked to recruit 10 players across age groups. Actual invitations were unknown; the summary contained 120 returned gameplay records from adults aged 25–80 years, all new to this study game and including the distributors' own records. Delivery included emailed instructions and prepared iPads, with keyboard-arrow and/or foot-pad controls. Aggregate engagement classifications were analysed descriptively. Clinical performance, self-appraisal and learning were not assessed in Stage 3A. Subsequent Stage 3B Pro development exports were examined separately as a developmental record.

Results: The Stage 3A summary classified 23 of 120 records (19.2%) as initial dropouts and 15 of these 23 (65.2%) as returned. Initial-dropout classifications were 0 of 38 (0.0%), 7 of 41 (17.1%) and 16 of 41 (39.0%) in the 25–44, 45–64 and 65–80 age bands. Unavailable applied classification rules and event timing limit interpretation.

Conclusions: The deployment supports a research direction in which abandonment is investigated in relation to self-appraisal and opportunities to learn from experience. It does not establish FIAS as a distinct mechanism, an effect of clinical decoupling, or adaptive-learning efficacy. Source-linked behavioural records and direct measures of appraisal and learning are needed to test these propositions.

Keywords: gamification; older adults; engagement; self-appraisal; coordination game; fall risk assessment; exploratory observational study; adaptive learning

Introduction

Background

A survey of 200 residents at a Guangzhou care facility, from a resident population of 280, found that 39% had used an installed fall risk assessment (FRA) system; among prior users, approximately 74% reported an adverse or unacceptable first experience and, of that subgroup, approximately 82% reported no subsequent return [1]. A subsequent eight-week deployment with 30 residents examined engagement with a gamified FRA at the clinical station. In that cohort, 21 of 30 (70.0%) completed three or fewer game cycles (attended play sessions at the station), and no re-engagement was observed among those 21 low-engagement participants during the eight-week window [2]. These earlier observations motivated a redesign of the way participants encountered the game.

The programme proposed failure-induced abandonment syndrome (FIAS) as a provisional label for non-return following an adverse, personally salient experience [1,2]. The central hypothesis concerns a self-cognitive challenge: a player may interpret performance in a clinically attached game as evidence about their own capability, making further participation personally threatening. FIAS is not a clinical diagnosis or a validated measurement construct. Technology-acceptance and gamification approaches remain relevant [3,4,5]; the senior technology acceptance model already includes self-efficacy and anxiety [4]. The present hypothesis therefore requires evidence of a specific appraisal process, not simply evidence of abandonment. Loss aversion and the endowment effect offer possible theoretical analogies [6,7,8], but neither these processes nor identity threat were measured here.

Gerontoludic design prioritises enjoyable play and the diversity of ageing [9], while affective game planning explicitly connects appraisal to meaningful interaction [10]. These perspectives support examining the meaning a player assigns to an experience. Usability, accessibility, motivation and ordinary attrition also remain possible explanations for non-return [11]. Clinical decoupling is a design strategy for investigating the appraisal hypothesis; it cannot by itself verify FIAS or exclude these alternatives.

Study Objectives

Stage 3A re-implemented the skiing-style coordination game as a separate, distributable HTML build, played away from the clinical station without the FRA assessment apparatus or clinical outputs; the Stage 2 game was itself a browser game, played at the clinical station. Its primary objective was to describe the available engagement classifications and their age distribution in this setting. Its conceptual objective was to examine how this decoupling could inform investigation of FIAS as a proposed self-appraisal mechanism. A further design objective concerned learning from experience: continued play could provide opportunities to adjust strategies and responses. Stage 3A did not measure such learning. Subsequent Stage 3B development explored alternative mechanisms for a future adaptive-learning evaluation and is discussed separately from the Stage 3A results.

Methods

Study Design and Setting

Stage 3A was an exploratory observational deployment conducted between April and July 2026, reported through aggregate tabulations of engagement classifications. Multimedia Appendix 1 maps the report to the STROBE checklist for cross-sectional studies [12] and identifies unavailable information. Recruitment used Hong Kong-based personal and community networks, including work

contacts, a charity community network, university support channels and medical-centre peers. Gameplay occurred away from the clinical FRA station. Supervision was not required. Some older players received help operating a prepared iPad from a research assistant, without the assistant playing on their behalf. The number and extent of assisted sessions were not recorded by participant. The therapist-supervised Stage 2 cohort [2] provides historical context and was not a concurrent control group.

Participants and Recruitment

We invited 20 distributors, acting as research assistants, and asked each to find about 10 players across age groups, giving a nominal recruitment target of about 200. The number of distributors who actively recruited, the number of people they actually invited and the number who began play without returning a file were not compiled. The target was a recruitment aim rather than a formal power calculation. We received 120 gameplay datasets represented in the aggregate summary. The investigator reports that the distributors, members of separate Hong Kong groups without prior knowledge of the gamified FRA, were eligible and also played, and that their own records are among the 120; their records cannot be separated from those of the players they recruited. Accordingly, 120 is the returned-record denominator; it is not an invitation count, and dividing it by the target of 200 would not provide a response rate.

Stage 3A players had not previously played this study game. This refers to study-specific familiarity, not general gaming or technology experience. Stage 3A used a different cohort from the Stage 2 residents. The summary records ages of 25–80 years: 38 records in the 25–44 band, 41 in the 45–64 band and 41 in the 65–80 band; 48 were recorded as men and 72 as women, and all as having high-school education or above. These demographic counts are retained as compiled. Distributor briefs addressed players aged 18–45 and players aged 45 and over; these brief categories differ from the reported analysis bands. Recruitment through distributors and onward forwarding constitutes a nonprobability network sample with possible selection and researcher influence. Participant-level linkage and duplicate checking are required to confirm the correspondence between returned files and unique participants.

Participant Information and Game Presentation

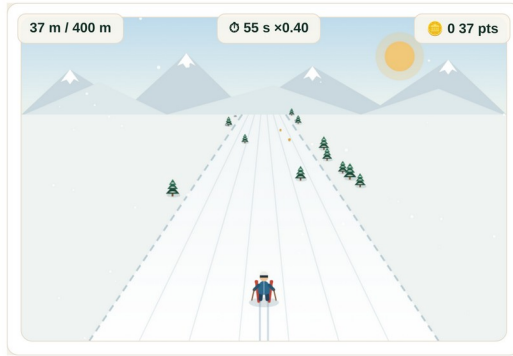
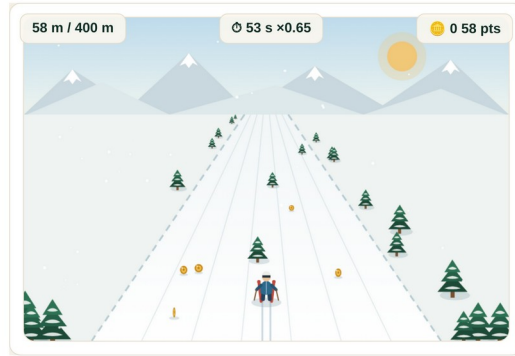
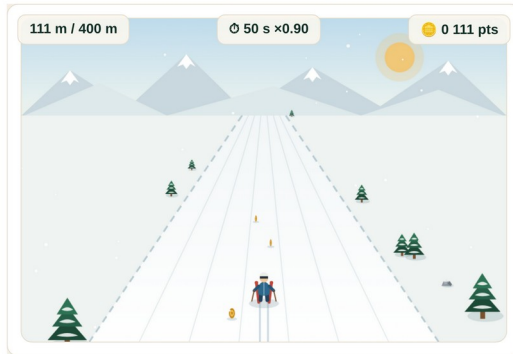
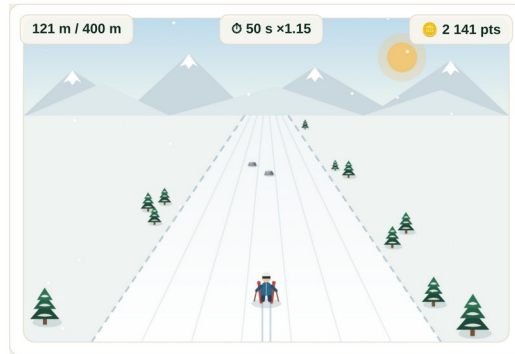
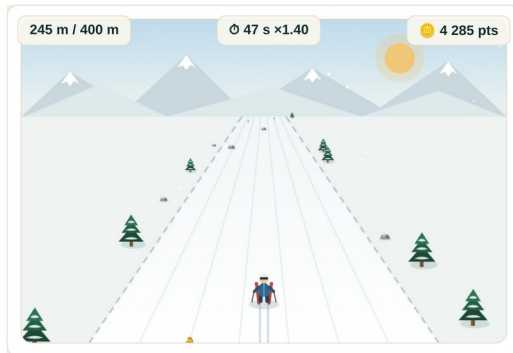
Distribution included remote email invitations with introductory information and prepared iPads with the game preloaded for some older players. We describe the participant invitation as a university-headed invitation to play a skiing game, introducing falls and balance training and disclosing the game's origin in an FRA platform. The available invitation document is a reconstruction received on 4 August 2026 whose heading gives the study period as “From March to April 2026”, whereas the research assistant reports that recruitment began on 3 April 2026; it also describes arrow-key or touchscreen controls, although neither Stage 3A build has touch controls, and the original version as distributed has not been linked to the study record. A distributor email described a coordination game and stated that the clinical context was being dropped. The original communications are needed to establish the health-related wording each participant received.

Clinical decoupling removed the pressure-panel assessment station, the pre-session clinical scan and participant-facing clinical scores or risk outputs. Stage 3A used keyboard-arrow and/or foot-pad controls, retained to keep the control demands closer to those of the Stage-2 station, where lateral weight shifts on a generic pressure mat placed on the FRA footplate were read by the game as left/right arrow keys. This intended similarity does not establish equivalent motor demands or

measurement validity. Device, location and assistance could vary, and recruitment could still mention the health-related origin. Decoupling therefore denotes removal of clinical apparatus and outputs within a broader deployment change; exposure to health-related framing was not experimentally controlled.

Game Build and Intended Procedure

Two Stage 3A builds are archived. A copy of the earlier build ('STAGE 3 RESEARCH BUILD v1.0'; 21,521 bytes; SHA-256 prefix 3200618c), distributed from April 2026, was supplied by the investigator on 30 September 2026 and is archived as received. The redesigned build, `Alpine_Coordination_Game_Stage3_v2.2.html` ('Stage 3 Short-Run Build v2.2'), is a 37,279-byte HTML file with SHA-256 prefix 42bc3df6; the archive identifies it as byte-identical to the `_v2` file. Runs use a distance goal or a time cap: 200 m or 30 seconds, and 400 m or 60 seconds. The zero-failure mode does not slow the player following a bump. The investigator reports that distributors received both the v2.2 build, first created on 13 July 2026, and the earlier v1.0 build, in which a run could end in a game over, and passed them on. Both builds open in zero-failure mode by default and carry a game-over mode that must be switched on. Returns did not record which build a participant played, so the number who played each is unknown. The investigator reports that implementation limitations with the earlier version led to this build and that the main period of play was postponed to late July 2026, so most returns, including returns after an initial dropout, fell within a few weeks before the summary was compiled in July 2026; the window for a return after the main period of play was therefore short, its length is not recorded, and a return on the earlier version could have occurred at any time after first play. Both builds support keyboard arrows and a joystick or foot pad through the browser Gamepad API; neither has touch controls. Prepared-iPad delivery therefore does not imply touchscreen input: the reported tablet-assisted sessions used an external keyboard. Participant-specific device and controller combinations are unavailable. Touchscreen input belongs to the later Stage 3B design. Figure 1 illustrates the archived Stage 3A design at staged difficulty settings.

Level 1 (difficulty $\times 0.40$)**Level 2 (difficulty $\times 0.65$)****Level 3 (difficulty $\times 0.90$)****Level 4 (difficulty $\times 1.15$)****Level 5 (difficulty $\times 1.40$)**

Difficulty rises from Level 1 to 5: obstacle spawn rate scales with the multiplier; stones grow more frequent relative to trees; boulder pairs appear from Level 3 upward. The skier is steered by lateral movement (keyboard arrows, controller or foot pad); coins reward forward play. Captured from the programme's archived game build.

Figure 1. Archived Stage 3A v2.2 build (Alpine_Coordination_Game_Stage3_v2.2.html) at staged difficulty multipliers; the build is identified from the screenshots themselves by palette, run display and boulder-pair feature. Steering uses keyboard arrows, a controller or foot pad. The interface omits clinical labels, clinical scores and risk outputs. These screenshots illustrate the archived v2.2 build and do not verify the settings or build used by every participant.

The distributor brief suggested five sessions, whereas the aggregate register describes initial dropout as stopping before five attempts. Whether these units meant separate visits or runs within one sitting is not established. The intended procedure is therefore distinguished from verified completion. Both Stage 3A builds include an optional request for language-model difficulty advice, falling back to a built-in rule agent that steps difficulty up above 80% obstacle avoidance and eases it below 40%. The distributed HTML files carry no embedded credential and browser cross-origin policy blocks direct model calls, so every Stage 3A session ran under the built-in rule agent and no language-model call operated in any Stage 3A session; the build saved nothing automatically, so no runtime log was kept.

Artificial intelligence (AI) assistance in software development is separate from an AI system operating during gameplay; no language-model effectiveness is claimed.

Outcomes and Source Records

The primary available quantities are the register's classifications of initial dropout, returned among initial dropouts, and non-return among initial dropouts. The descriptive initial-dropout label is "dropped out before completing 5 attempts". The investigator reports that the classifications applied a failure-then-stop rule: a player who stopped after a failure, including a game over in the earlier version, was classified as an initial dropout; one who later played again, on either build and at any time before the July 2026 compilation, as returned, regardless of how long or how often they then played; and one who never played again as a non-return. For older players, the failure and the stop were observed by an assistant; for others, they were inferred from reported results or from not playing again after the first game, so some initial dropouts have no recorded failure. Exporting results from the archived build was optional, so returns arrived over several weeks. The source records behind each classification are not linked to the summary. Dates of first play, disengagement and return, a disengagement checkpoint, reminders and assistance are unavailable in the compiled record. Consequently, these counts cannot distinguish immediate replay from later re-engagement or prompted from unprompted return. The complement of the eight non-return classifications is 112 records, which does not establish five-attempt completion or activity at an observation-window endpoint and is not a retention outcome.

Two questionnaire-related counts are retained as positive-coded responses in the aggregate summary. The summary labels the items as liking the game as a reflection of eye-leg coordination and using it for regular health training if accessible; the retrospective invitation instead describes a coordination challenge and willingness to play regularly. The investigator reports that the questionnaire contained these two items and demographic questions, used a five-point scale on which 4 or 5 was coded positive, and was answered by players immediately after they chose to stop playing. The questionnaire as sent, including its language, has not been linked to the study record, and the difference between the summary labels and the invitation wording is unresolved. These counts are not treated as validated measures of coordination, intrinsic motivation, intention or the success of a framing manipulation.

All Stage 3A numerical results reported here originate in the Preliminary_Summary sheet of the Stage 3 data logbook. Its participant-level sheets do not contain the source-linked observations needed to regenerate the summary or verify unique participant counts. No participant names or email addresses were collected. With the first build, results and stopping were recorded by hand by assistants who were not always beside the player; later returns arrived by email in mixed formats. The returned results and questionnaire responses were compiled into the summary in July 2026 and were not retained as separate records, so record-level exclusions, duplicate checks and missingness cannot be reconstructed. No imputation or participant-level adjustment was performed. Records from subsequent development exercises are not added to the Stage 3A denominator.

Statistical Analysis

Counts and percentages are reported with explicit denominators. Wilson score 95% confidence intervals use the reported binomial counts. An exploratory chi-square test examines age-band differences in initial-dropout classifications, and a Fisher exact test compares the youngest and oldest bands. These unadjusted comparisons were selected after tabulation; they were not prespecified and

no multiplicity adjustment was applied. Analyses use the available aggregate age bands, without participant-level adjustment for device, assistance, education, prior technology experience or distributor clustering. Intervals and tests are conditional on the reported counts and assumed independence; they do not resolve selection or classification uncertainty. No causal contrast with Stage 2, time-to-event analysis, paired behaviour–intention analysis, sensitivity analysis or learning-effect estimate was calculated.

Stage 3B Pro Development Records

Twelve Pro exports from 19–20 September 2026 (UTC; builds 4.1 and 4.1.1), covering seven participant codes, were examined retrospectively as a separate development exercise. Ten, covering five codes, were summarised in two dated reports of 20 September 2026 (a batch research review and a learning-measurement addendum); two further exports, which were received by different routes and reached the record on 29 September 2026, were added from their raw files. The software labels every export as a researcher self-test; the investigator reports that the codes were played by volunteers and their families or friends recruited for selected age groups. Each export records an in-game consent tap, and the exports do not establish whether more than one person played under a code. Counts therefore refer to codes, runs and software-defined visits, without assuming verified individual attribution of every run or independent participants for each visit.

Repeated cumulative exports were reconciled by code and run identifier; duplicate practice-detail rows in CSV exports were not counted again. Adaptation decisions were checked against linked performance and feedback records and the next run’s recorded settings. Complete within-visit opening and closing checks were described using obstacle clearance, calculated as $100 \times (\text{obstacles faced} - \text{obstacles hit}) / \text{obstacles faced}$. These descriptive checks do not estimate a learning effect or establish equivalence of different courses.

Ethical Considerations

Participation was voluntary and unpaid. According to the retained account of the invitation and distribution emails, participants were informed of the purpose of the game and the accompanying questionnaire and of the use that would be made of replies, that participation was voluntary and unpaid, and that they could decline by not taking part. Returning the results file and questionnaire responses constituted implied informed consent; no written consent instrument was administered. No personal names were requested at enrolment. The returned files were not retained as separate records after compilation, so implied consent is documented by the invitation wording and the aggregate record rather than by individual returns. Acceptance of implied consent on this documentation is a matter for the Departmental Research Committee; no determination on it was on file at 30 September 2026. Stage 3A did not conduct a clinical health-performance assessment.

Stage 3A recruitment began on 3 April 2026 after the supervisor’s confirmation to proceed. According to the research assistant who handled collection, most players took part after the recruitment deadline, and returns from older players in particular arrived late because reaching them required communication and arrangements; the last returns were compiled and processed in July 2026. An administrative misunderstanding between the researcher and the administration office delayed the formal application; this account has not been linked to a contemporaneous authorisation record. The programme’s human-subjects ethics application, HSEARS20260805008, was submitted and approved on 6 August 2026 by the Departmental Research Committee of the Department of Industrial and Systems Engineering on behalf of The Hong Kong Polytechnic University Institutional

Review Board, with a stated approval period of 16 June 2024 to 30 October 2026. By the first author's account of the application as submitted, which is held by the first author, it described the procedures of Stages 1–3 as already conducted and asked the Committee to review and characterise them. Thus, the formal decision followed Stage 3A collection. The supervisor's confirmation and the later formal decision are distinct. The approval memo is a one-page project-level approval for the stated period; it does not itemise determinations on the earlier procedures or their consent arrangements, and whether the approval covers the Stage 3A procedures and the implied-consent arrangement as conducted is not determined in the record. On 3 October 2026 the principal investigator of the approved application (CFC) advised the first author that no further request to the committee was needed (by the first author's report of the same date); no itemised written determination is on file and none is sought, and the chronology and the record are reported as they stand.

The Pro exports retain the software label “researcher-self-test”, with participant-release or approval fields set to false and ethics references absent or not asserted. The investigator's account of volunteer use is reported separately from those settings. The Pro build's own record states that no participant may use it until the approval is confirmed to cover a variant able to send player reports to a third-party model service and the supervisor approves; by the investigator's account of 30 September 2026 the codes were nonetheless played by volunteers. Whether the Stage 3B collection, including this use, falls within the approved scope (HSEARS20260805008, approval period to 30 October 2026) or requires an amendment is recorded as an open item of the programme's ethics record: no clarification or amendment is on file, the supervisor's sign-off of the Stage 3B protocol is not recorded as closed, and following the principal investigator's advice of 3 October 2026 no further request to the committee is being made. The software flags alone do not establish the ethics status of the use.

Results

Compiled Classification Counts

Of the 120 records in the summary, 23 (19.2%) carried an initial-dropout classification. Fifteen of those 23 (65.2%; 95% CI 44.9%–81.2%) also carried a returned classification and eight (6.7% of all records) a non-return classification. The remaining 97 records are the arithmetic complement of the initial-dropout count. Table 1 presents these quantities without treating the labels as a verified sequence of behavioural events.

Table 1. Compiled Stage 3A classifications with explicit denominators and conditional Wilson 95% confidence intervals.

Classification	Count / denominator	Percentage	95% CI in %
Initial-dropout label	23 / 120	19.2	13.1 to 27.1
Returned label among initial dropouts	15 / 23	65.2	44.9 to 81.2
Non-return label among initial dropouts	8 / 23	34.8	18.8 to 55.1
Non-return label in the full summary	8 / 120	6.7	3.4 to 12.6
Complement of the initial-dropout count	97 / 120	80.8	72.9 to 86.9

Intervals are conditional on the compiled counts and a binomial independence assumption; they do not account for selection, distributor clustering or uncertain classification. The complement of initial dropout does not establish protocol completion.

Initial-dropout classifications increased across the age bands: 0 of 38 (0.0%), 7 of 41 (17.1%) and 16 of 41 (39.0%). The three-band chi-square test yielded $\chi^2(2)=19.56$ ($P<.001$), and the youngest-versus-oldest Fisher exact test also yielded $P<.001$. These are exploratory, unadjusted associations between age band and compiled labels. Among initial-dropout records, five of seven in the 45–64 band and ten of sixteen in the 65–80 band also carried a returned classification. The youngest band had no initial-dropout records, so its conditional returned percentage is undefined (Table 2; Figures 2 and 3).

Table 2. Age distribution of the compiled classifications.

Age band in years	Records, n	Initial dropout, n (%)	Returned, n / dropout n	Non-return, n (%)
25-44	38	0 (0.0)	Not applicable	0 (0.0)
45-64	41	7 (17.1)	5 / 7	2 (4.9)
65-80	41	16 (39.0)	10 / 16	6 (14.6)
Total	120	23 (19.2)	15 / 23	8 (6.7)

Percentages in the initial-dropout and non-return columns use the age-band denominator. Returned classifications are a subset of initial dropouts. For ages 65–80, 10/16 returned classifications correspond to 62.5% (95% CI 38.6%–81.5%); 6/41 non-return classifications correspond to 14.6% (95% CI 6.9%–28.4%). The labels do not establish an observation-window endpoint or unprompted return.

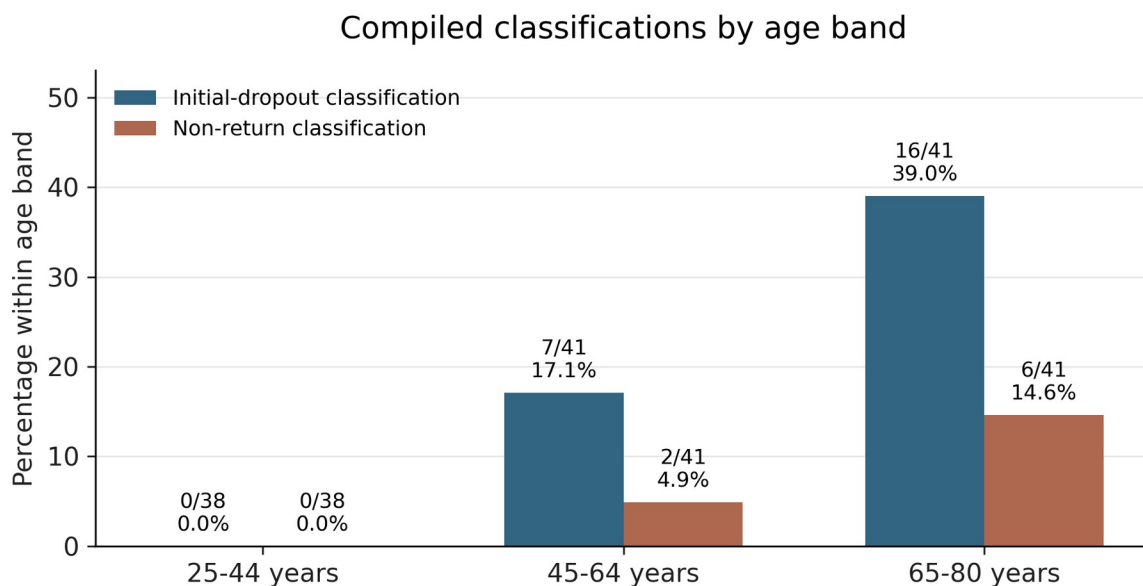


Figure 2. Initial-dropout and non-return classifications as percentages of each age band. Non-return classifications are a subset of initial-dropout classifications. The labels do not establish a defined end-of-window outcome.

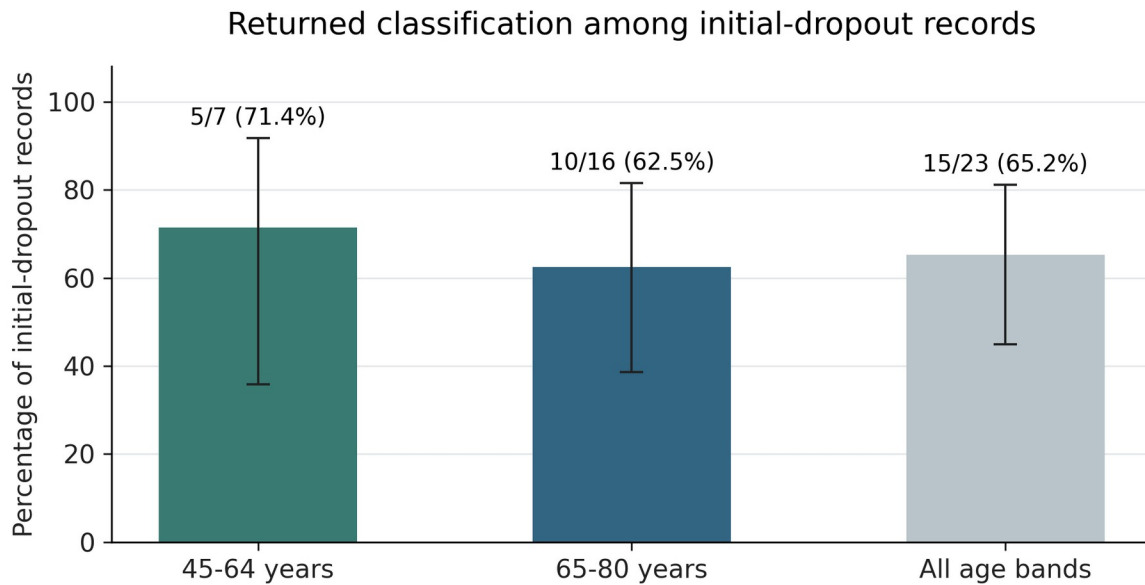


Figure 3. Returned classifications among initial-dropout records. Error bars are conditional Wilson 95% intervals from the reported counts. The 25–44-year band had no initial-dropout records, so its conditional percentage is undefined. Stage 2 is not plotted as a baseline or control.

Questionnaire Coding in the Aggregate Summary

The summary records 86 of 120 responses (71.7%; 95% CI 63.0%–79.0%) as positive on the coordination-related item and 76 of 120 (63.3%; 95% CI 54.4%–71.4%) as positive on the regular-use-related item. The latter counts are 25 of 38, 28 of 41 and 23 of 41 across ascending age bands (Table 3). These are coding frequencies whose administered form has not been linked to the study record. The coordination-item age breakdown is unavailable. The summary provides no item-level missing-data flags. No “commitment gap” is inferred because the behavioural classifications and questionnaire items do not establish comparable or paired constructs.

Table 3. Positive-coded questionnaire responses reported in the aggregate summary.

Summary item	25-44 years	45-64 years	65-80 years	All records
Coordination-related item	Not available	Not available	Not available	86/120 (71.7%)
Regular-use-related item	25/38 (65.8%)	28/41 (68.3%)	23/41 (56.1%)	76/120 (63.3%)

Item labels are descriptive identifiers for the compiled counts. They do not replace the original administered wording, response scale or scoring threshold.

Stage 3B Pro Developmental Findings

After reconciliation, the exports represented 53 distinct runs in 16 software-defined visits under seven codes: 21 practice runs, ten warm-ups and 22 performance checks. Repeated run records showed no substantive conflicts. Five codes recorded touch input only, one recorded keyboard input, and one recorded touch input plus a stopped run with no input. Prior-play responses do not identify a paired follow-up cohort from Stage 3A. Software visits do not necessarily represent separate days or independent players.

Twelve decisions could be linked to the preceding performance and player-feedback records and to the settings recorded for the next run: nine were final local-rule decisions and three were player overrides of the proposed setting. No cloud-model request was recorded. Twenty-one practice-feedback records comprised 12 answered and nine explicitly skipped responses. This documents implementation of feedback-informed adaptation, without establishing a benefit of adaptation or of a language model.

Six complete opening–closing performance-check pairs occurred under five codes, two of them under one code. Obstacle clearance changed from 100% to 90%, 95% to 100%, 95% to 90%, 100% to 100%, 80% to 90% and 80% to 90%, respectively (−10, +5, −5, 0, +10 and +10 percentage points; three increases, two decreases and one tie; mean +1.7 and median +2.5 percentage points), and two opening checks were already at the 100% ceiling. Each check involved 20 obstacles with a nominal 30-second duration, but course versions differed and equivalence was not established. These mixed immediate changes, without a comparator or delayed unassisted assessment, do not demonstrate durable learning.

Discussion

Principal Findings

This study places self-appraisal at the centre of a question raised by abandonment of a clinically attached coordination game. Stage 3A implemented one way to investigate that question: make the game available outside the FRA apparatus and remove clinical outputs while retaining coordination-oriented play. The summary contained returned classifications for 15 of 23 initial-dropout records, including 10 of 16 in the oldest band, alongside an age gradient in initial-dropout labels. These observations motivate further investigation of how players interpret and revisit challenging experiences. They do not demonstrate a psychological mechanism or an observed disengagement–return sequence.

Comparison With Prior Work

The comparison with the earlier clinical deployment is contextual. Stage 3A involved different participants, a wider age range, remote distribution by email and play away from the station, variable settings and assistance, and different equipment. Keyboard-arrow and/or foot-pad control was intended to retain aspects of the earlier interaction, in which lateral weight shifts on a pressure mat were read by the game as arrow keys, but does not make the deployments equivalent. Stage 3A players were new to this study game, whereas the Stage 2 residents had experience of the clinically attached system. Health-related framing at recruitment is incompletely documented. Differences between these deployments cannot identify an isolated effect of decoupling.

FIAS proposes that a personally adverse interpretation of an assessment experience contributes to subsequent withdrawal. This is a hypothesis about self-appraisal, rather than a diagnosis inferred from low use. A useful test would measure perceived failure or capability threat after comparable experiences and link those measures prospectively to a defined return outcome. The two Stage 3A builds, both of which open in zero-failure mode by default and carry a game-over mode that must be switched on, the unrecorded build each player received, absence of direct appraisal measures and unavailable event chronology prevent such a test. A bump is not necessarily experienced as failure. The earlier observation that three of the six residents in the single-session non-return cluster (F4) of [2] recorded a positive FRA-index change in that single session and did not return during the

observation window, one of the three being the record that failed verification and was retained under a sensitivity analysis [2], illustrates why measured performance and willingness to continue may diverge; it does not establish the cause of their withdrawal.

Existing acceptance and appraisal-based design frameworks already recognise self-efficacy, anxiety, meaning and personal interpretation [4,9,10]. The proposed contribution is therefore a specific, testable account of abandonment in the FRA context and a decoupling strategy for investigating it. The present design cannot exclude usability, game mechanics, access, fatigue or social support as contributing explanations. Age-band differences also do not establish uniform preferences or capabilities among older adults.

Game Achievement and Learning From Experience

Three outcomes require distinct interpretation. Game achievement concerns completion, distance, points or other performance within a game. Clinical FRA performance requires the relevant assessment apparatus and a validated relation between its measurements and the clinical construct. Learning from experience concerns changes in skill, strategy or appraisal arising through practice and feedback. Stage 3A retained game achievement recording but provided no corresponding clinical FRA measurement; this report analyses aggregate engagement classifications rather than achievement trajectories. Removing clinical measurement changes what can be inferred from the recorded game data.

Continued participation creates an opportunity for learning but is not evidence that learning occurred. Immediate improvement may reflect familiarity, assistance or a change in difficulty. Durable learning requires suitable retention or transfer assessments [13]. Self-efficacy and stereotype-threat literature can guide hypotheses about how feedback is interpreted [14,15,16], but these mechanisms were not measured here. If clinically valid risk information is reintroduced, it should be communicated accurately with appropriate support. Game scores should not be presented as a diagnosis or a judgement of personal worth.

Stage 3B and Adaptive Learning Development

Stage 3B was a subsequent pre-specified (repository-dated) 2×2 component study, reported here as a descriptive record, intended to examine whether alternative mechanisms could support adaptation and the measurement of learning from gameplay. The design retained a similar skiing aesthetic while changing mechanisms with assistance from Anthropic’s Claude Code and introducing touchscreen input. The original component protocol, dated 8–9 August 2026, contrasted adaptive and static presentation and zero-failure and game-over modes; it did not manipulate clinical framing. Using Claude Code to help develop the software does not establish that a language model adapted play in real time.

Some Stage 3B players had previously played the study game, and its build asked about prior play at enrolment. The original component phase has yielded insufficient comparable data. The record comprises one verified return of 15 August 2026 on the deployed 9 August build; five returns received on 15 September 2026 that were excluded because they were played on the 13 August revision rather than the deployed build, were researcher-solicited plays made within about twenty minutes on one device copy after a re-send, and all fell in the adaptive arm; and five clean-run returns from the frozen study edition of 15 September 2026, played on 16 and 29 September 2026, three of them single sessions, within a clean-run window that runs to 15 October 2026; one further file (code BYXXMT), with stamps dated 13 August 2026, remained under verification at the cut-off. The

record reported here is closed at 30 September 2026, and later returns are archived but not reported. The September Pro exports are a separate developmental record: they document operation of a rule-based adaptation process, without testing its learning benefit. Seven of 16 software visits contained no practice, and three of the nine visits with practice lacked a closing check. Prior exposure, input modality, changing builds and unresolved player attribution limit interpretation. Neither the original phase nor the Pro records are pooled with Stage 3A.

Limitations

The unknown invitation denominator and nonprobability recruitment limit generalisation beyond the returned-record sample. The target of 200 does not establish how many people were approached or explain why files were not returned. Selecting only file-returning players may favour those willing and able to engage; the direction and magnitude of the overall bias cannot be estimated from the available record. A source-linked participant reconstruction is needed to verify unique participants, exclusions and variable-level missingness. Unlinked source records, the unrecorded build version and missing event timing limit the behavioural meaning of the counts. The questionnaire as administered also requires linkage to the record.

Device combinations, assistance and distributor effects could influence participation but cannot be linked to individual outcomes; the 120 records include the distributors' own play. Changes across stages involve input modality, setting, mechanisms and prior experience, precluding attribution to any one change. No clinical outcome, identity-threat mechanism, durable learning outcome or language-model benefit was measured. The formal ethics decision followed Stage 3A collection; the chronology is reported as it stands. Whether the approval covers the earlier procedures and their consent arrangements as conducted remains an open item of the ethics record: no itemised written determination is on file, and, following the Principal Investigator's advice of 3 October 2026 that no further request to the Committee was needed, none is sought.

Future Evaluation

A test of the FIAS hypothesis could compare clinical and non-clinical presentations within the same recruitment population while holding equipment, mechanics and assistance constant. It should record exposure to framing, measure appraisal of capability and perceived failure, and prospectively define disengagement, prompting and a return window. Measures of usability, access and technology acceptance would help assess competing or interacting explanations. Prior play should be recorded and accounted for, with version identifiers, participant links and timestamps preserved. If fear of falling is relevant, a suitable instrument such as the Falls Efficacy Scale-International [17] could be included; it would not substitute for measuring the proposed self-appraisal mechanism and was not administered here.

A future adaptive-learning study should specify the target skill and what the adaptation changes, compare appropriate conditions using equivalent assessments, and measure delayed unassisted retention or transfer. Repeated participation by the same person must be identifiable and analysed accordingly, with separate identifiers for invited players. Game achievement, perceived improvement and durable learning should remain separate outcomes. The Pro logs support the technical implementation of feedback-informed adaptation; feasibility of a complete learning evaluation and its efficacy remain unestablished. Feedback capture should also be built into the game. The Stage 3A builds had no in-game consent step or questionnaire, so responses were collected outside the game and returns arrived in mixed formats; the Stage 3B builds recorded consent, asked questions

immediately after play and exported runs, condition and answers in one structured file, which made each return checkable but did not secure returns, because players still had to send the file. Any further collection requires the applicable prospective ethics determination and prespecified hypotheses and endpoints.

Conclusions

Decoupling a coordination game from FRA apparatus and outputs provides a design approach for investigating whether self-appraisal contributes to abandonment. The Stage 3A aggregate record describes age-related engagement classifications, but cannot establish FIAS, verify an effect of decoupling or demonstrate learning. Stage 3B extends the research direction toward adaptive learning with similar aesthetics and different mechanisms. Its Pro logs record rule-based adaptation, while its learning measurement does not reach the adaptive-learning construct and no inference about learning is drawn. The contribution is an exploratory account linking the appraisal question to a concrete design change and identifying the behavioural and learning measurements needed to test it.

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Data Availability

The aggregate Stage 3A data logbook and both archived Stage 3A game builds (a copy of the earlier v1.0 build, received on 30 September 2026 and archived as received, and the v2.2 build created on 13 July 2026) are retained with the first author's thesis records. Distributor invitations are retained. The returned results and questionnaire responses were compiled into the aggregate summary and were not retained as separate records, so no participant-level Stage 3A dataset exists. The twelve Stage 3B Pro source exports and the dated batch review of 20 September 2026 (two reports) are archived as received with SHA-256 checksums in the first author's thesis repository, which also holds the Stage 3B component-phase returns (codes only: one verified return of 15 August 2026, five excluded and five clean-run) and the frozen study edition with its checksums; the derived workbooks and extraction scripts named in the reports are not deposited; the exports' software metadata and the investigator's account of who played are distinguished in reporting. Requests for access should be directed to the corresponding author and require review of de-identification, participant consent and institutional conditions.

Author Contributions

SKTT led the research programme, deployment, compilation of records and manuscript preparation. CFC supervised the research programme and reviewed the preceding manuscript version.

Conflicts of Interest

The authors declare no competing financial interests. The Stage-2 study was conducted with the residential care provider operating that facility; the provider had no role in the present Stage-3 study. Participants were recruited through the authors' networks.

Use of Generative Artificial Intelligence

The study game was created with generative-AI coding tools under the first author's instruction and design framework: the Stage 2 game, whose skiing-game concept the Stage 3A build retained, was generated and corrected through OpenAI's ChatGPT, and Anthropic's Claude Code was used from 2025 for the later Stage 2 build generations, the separately built Stage 3A HTML builds and the subsequent Stage 3B mechanism development. OpenAI's ChatGPT and Anthropic's Claude assisted manuscript revision, consistency review and plotting-code preparation for the 27 September 2026 version; ChatGPT assisted the revision and descriptive review of Pro records on 29 September 2026; Anthropic's Claude assisted consistency review of subsequent revisions against the first author's thesis records. Figures 2 and 3 were generated deterministically from the reported aggregate counts using plotting software. No participant observations were generated. The authors retain responsibility for the accuracy of the text, references, analyses and figures. Development or writing assistance does not establish operation or effectiveness of a language model during gameplay.

Related Publications

The programme's Stage 1 conference paper [1], presented in March 2026, and Stage 2 companion manuscript [2], under review at the Journal of Information and Knowledge Management, concern separate study records. Their results are cited as background and are not new Stage 3A findings. The primary quantitative results use the Stage 3A summary compiled in July 2026. September Stage 3B Pro logs are reported separately as developmental evidence and are not pooled with Stage 3A or the original Stage 3B component-phase records.

Abbreviations

AI: artificial intelligence
API: application programming interface
CI: confidence interval
CSV: comma-separated values
FIAS: failure-induced abandonment syndrome
FRA: fall risk assessment
HTML: Hypertext Markup Language
SHA-256: Secure Hash Algorithm 256-bit
STROBE: Strengthening the Reporting of Observational Studies in Epidemiology
UTC: Coordinated Universal Time

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